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### Analysis of Data Centers' Water Consumption in the US

With the rise of Industrial Revolution 4.0 in the 2020s, particularly following the launch of generative AI technologies such as ChatGPT, the demand for large-scale data centers has skyrocketed. These data centers primarily host servers—hardware devices that store data and run a wide range of software systems, including cloud computing services and online video games. Through extensive transcontinental optical fiber networks, these servers connect billions of users worldwide to digital services and platforms. While the Internet boom of the 1990s and 2000s initially drove the need for data centers for data storage, making information unprecedentedly accessible, the new AI revolution has created additional demand for data centers to store and process massive amounts of data, enabling CPUs and GPUs to deliver instant responses to user queries. Currently, the gigantic warehouse-like data centers are primarily operated by the big three cloud vendors — Amazon Web Services, Google Cloud Platform, and Microsoft Azure. Despite the conveniences these facilities provide, their significant environmental consequences are often overshadowed by the perceived benefits. These impacts can be categorized into three main areas: carbon footprint (from electricity generation), land footprint, and water footprint. This research paper analyzes water consumption in data centers, examines how these facilities use water for operations, evaluates their environmental impacts, and identifies feasible solutions to improve water efficiency.

According to George and colleagues, worldwide water footprints by sectors are dominated by agriculture (70%) and thermal power plants (15%) (93). Although data centers currently account for only 2% of global water use, their rapid growth is likely to become increasingly significant in the coming decades. Data indicates that the water footprint of data centers increased more than threefold over a decade, from approximately 21.2 billion liters in 2014 to 66 billion liters in 2023 (N. Lei et al. 2). Data centers consume a substantial amount of water directly to cool their power-intensive infrastructure, due to water's accessibility and high heat capacity. Additionally, a considerable volume of water is also consumed indirectly through electricity generation (N. Lei et al. 1).

Most servers are designed to operate safely at ambient temperatures within 45°C. However, the ambient temperature range for data centers is typically 18-27°C for optimal server efficiency (M. Azarifar et al. 3). As temperature increases, equipment failure rates also increase, though not necessarily linearly (Mytton 3). Data center cooling is primarily achieved through air-cooled or liquid-cooled systems. In air-cooled data centers, air-conditioning units circulate chilled air into the under-floor plenum, which is then passed into the room through the perforated tiles and circulated by fans. As the air heats up, it will eventually be transferred to outdoor air via a condenser, dry cooler, or cooling tower. In this method, a large amount of water is needed for evaporative cooling, where water absorbs heat from the hot air and evaporates, producing cool air again. A small 1MW data center using this cooling approach can use around 25.5 million liters of water per year. Air cooling has traditionally been the preferable method because it allows control over the humidity and air quality throughout the entire room (M. Azarifar et al. 3). However, air cooling methods faces its limitation: as data centers scale up, and more advanced and power-intensive devices are introduced, the heat generation at the rack level can easily

surpass 100kW—roughly enough to power 5-10 houses, making traditional air cooling method ineffective to provide sufficient cooling within optimal range for microprocessors to operate (M. Azarifar et al. 5). Consequently, liquid-cooling systems have been increasingly substituting air cooling systems in modern and large-scale cloud data centers to enhance heat dissipation. A notable method is the rear-door heat exchanger (RDHx), which circulates chilled water through a coil at the back of the servers to facilitate heat exchange as hot air is expelled (M. Azarifar et al. 5).

PUE (Power Usage Effectiveness) and WUE (Water Usage Effectiveness) are the two commonly used metrics for measuring power and water consumption in data centers. Both are influenced simultaneously by cooling system technologies, climate conditions, and operating efficiency practices (N. Lei et al. 3). WUE can be further divided into two categories: WUE-site (water used on-site) and WUE-source (water used to generate electricity). Research conducted by Nuo Lei and colleagues indicates that data centers employing water-cooled chillers have relatively higher WUE-site values; conversely, data centers with air-cooled chillers have relatively lower WUE-site values but higher PUE values (5). This phenomenon implies that data center administrators must make an optimal trade-off between consuming more natural cooling water vs using electricity to produce chill water. Nonetheless, colder climates and high operational efficiency still help lower the PUE and WUE-site scores. Traditional data centers have reported reducing PUE from 2.23 in 2010 to 1.93 in 2020. In contrast, large cloud data centers report PUEs ranging from 1.25 to 1.1, demonstrating better power efficiency (Mutton 2).

Even though higher PUEs generally correlate with lower WUE-site, they still lead to higher WUE-source because thermal power plants, such as coal, natural gas, and nuclear, require

a greater volume of water to cool themselves. A sensitivity analysis conducted by Nuoai Lei and colleagues found that the water consumption factor (WSF)—the amount of water used per unit of electricity generated—was the second most important factor in determining water use per workload in data centers, as shown in Figure 1 (10). This result indicates that powering the data center with a low-WCF electricity source can significantly reduce overall water use in the data center. WCFs is also a more general approach that is not constrained by the cooling technology implementation, but rather the availability of electric grid options. (N. Lei et al. 10). Solar and wind energy are ideal in this situation as they do not require any steam to spin the water-intensive turbine. On the other hand, server efficiency quantile ranked first in relative importance for overall water efficiency (N. Lei et al. 10), indicating that installing power-efficient servers is the most effective way to minimize water usage. These servers maximize the electrical energy to "do work" before it is lost as heat into the surrounding environment.

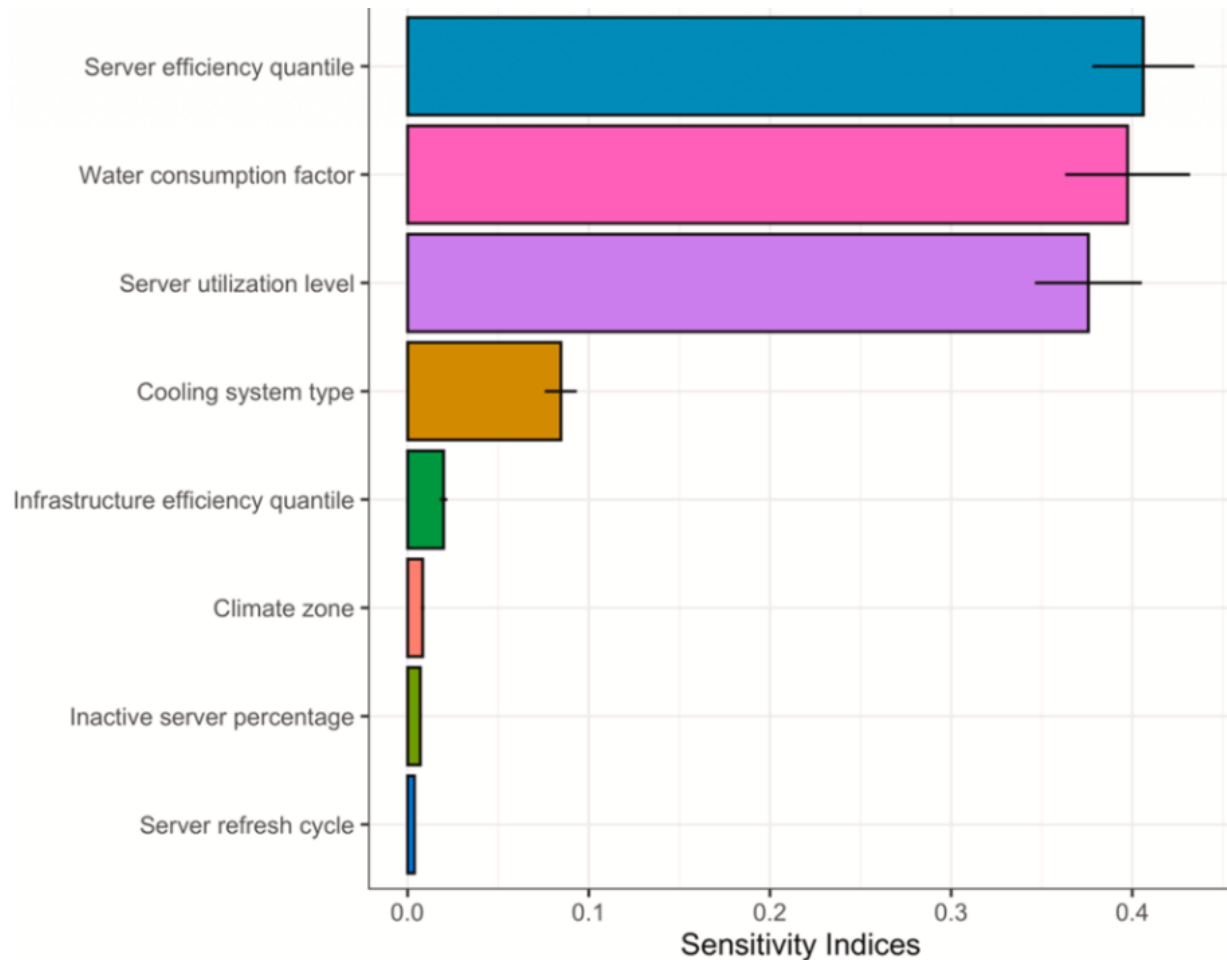


Figure 1: Importance ranking for all the determinants of the overall water use per workload. Screenshot from Lei, Nuo, et al., "The Water Use of Data Center Workloads: A Review and Assessment of Key Determinants," *Resources, Conservation and Recycling*, vol. 219, 2025, p. 10, article 108310.

When a single company administers a data center, it allows the company greater flexibility to explore alternative water sources. The big three cloud vendors have been experimenting with innovative approaches to reduce their power and water consumption. For example, Google's Hamina data center in Finland has been pumping seawater into the facility to cool its circulating hot water since it opened in 2011 (Mytton 11). Amazon claims to be the first data center operator approved to use recycled water effectively for direct evaporative cooling

(Mytton 12). Microsoft even undertook a bold attempt called Project Natick to tackle the challenge by submerging a sealed data center. This experiment is carried out by harnessing the cooling from ambient sea temperature and electricity from local renewable sources (Mytton 12).

There are many ways to cut down both power and water consumption simultaneously. Google has shown that shifting from cooling the entire room to focused cooling on a row of servers can improve energy savings by up to 29% (Mytton 4). Harnessing AI to make hourly adjustments based on predicted weather, ambient temperature, and other environmental factors can also yield an overall energy and water saving of up to 40% (Mytton 4). Besides that, increasing the efficiency of ICT equipment and data center infrastructure can significantly reduce the need for cooling for the same output (N. Lei et al. 11). Using renewable energy sources like solar and wind is also highly beneficial, as it contributes to lower carbon and water footprints. Lastly, managing data storage in an energy-efficient way can yield an observable reduction in overall resource usage over the life term of data storage, as data storage still accounts for a large fraction of energy consumption. According to D. Al Kez and colleagues, data structure can be classified into three categories: critical data (required information to operate firms), redundant data (semi-structured data that may no longer be used), and dark data (stored unstructured data no longer used for generating insights for decision making) (4). Due to how cheap hard-disk drives are and how valuable data is nowadays, unused data are still preferred to be stored electronically when businesses making their cost-benefit analysis, even though less than 1% of it is ever accessed again (D. Al Kez et al. 5). This results in a rapid expansion of redundant and dark data (86% of all data storage), which causes a significant amount of mostly non-renewable energy wasted just to run the storage devices and associated storage management systems (D. Al Kez et al. 6). Progressive firms who are committed to improving their sustainability are taking

actions to replace hard-disk drive with modern tape storage for dark data storage which is a viable technique to reduce their resource consumption in the long run (D. Al Kez et al. 7).

In conclusion, this research explores total water consumption in data centers, different cooling methods, the relationships between PUE and WUE and the environment, and ways to improve data centers' water efficiency. As industrialization continues to scale up and more resources are used, water — the most abundant resource on Earth — is being used increasingly aggressively. The scarcity of this "most abundant resource" has become a looming concern in the 21st century. Even though the rise of AI and data centers provides significant convenience to society, this overattachment to convenience may inadvertently impede society's long-term vision, leading to a sustainability bottleneck due to its "short-term pain, long-term benefit" nature. Even though people have become much more aware of environmental issues, environmental conservation is showing no signs of improvement as the hype around industrialization 4.0 continues to dominate the headlines. Solving the environment problems can never be done by one party alone, but by all. Hence, government and data center companies should work collectively to set clear environmental goals and foster a sustainability-focused dynamic within society. By internalizing the environmental externality cost into the profit calculation, society can learn to value the actual cost of using the products. Furthermore, at the educational level, instead of only teaching students how to maximize the benefits of AI, schools should also raise awareness of the environmental pitfalls of AI to produce a younger generation committed to sustainability and broader social benefits in the future.

## Works Cited

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